AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 24 · STUDENT LESSON

Privacy by Choosing Less

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

State the purpose. Collect less. Plan the whole information journey.

Redesign a fictional form, trace collection and disposal, and test whether a hidden name actually prevents identification.

Keep one habit: Justify each field → restrict access and sharing → name a responsible end point.

MAIN LESSON · SECTION 1

What information does this job actually need?

> Begin with the purpose, then justify each field.

Start with a concrete decision

Before adding a field to a form, explain the job it helps accomplish. A topic-counting task needs topic choices; extra names, photographs and addresses create information to protect without improving that count.

Recall what a calculation needs

A report needs the number of correct answers out of 20 fictional test cards. Does that calculation require a student’s name or home address? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can justify which information a stated task needs and remove unnecessary fields.
- I can distinguish reducing **personal data** from hiding a name.
- I can design a simple access, **retention** and deletion plan for a classroom activity.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

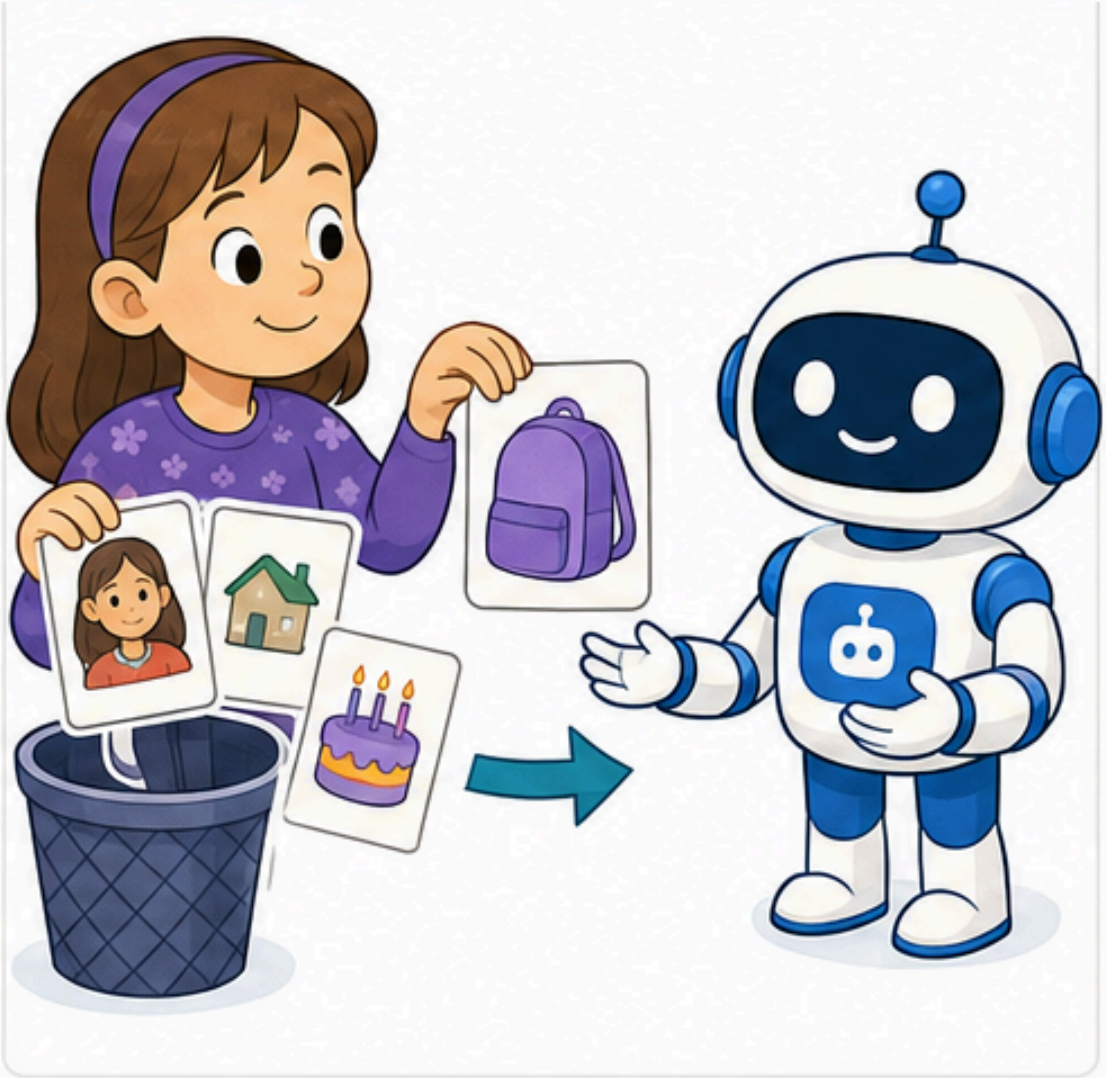
Remember before reading

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

A useful system should collect only information it truly needs.
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LEARNING TARGET A book recommender requests genre, maximum pages, home address and a selfie. Keep only fields needed for genre and length matching. Explain one consequence of storing extra fields.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can collect only information necessary for a stated task and explain coded-record limits.

MAIN LESSON · SECTION 2

Redesign the fictional workshop vote

> Collect only the topic choice for a topic-counting task.

Define privacy and the purpose

Privacy includes having appropriate boundaries around information about people. A useful first question is: what exact job are we trying to do? A second is: what is the least information that can do that job?

Collecting extra details just because a form has room for them creates **more information** to protect.

Read the fictional voting procedure

Our fictional class wants to choose one of three workshop topics for Friday: drawing, music or coding. It needs a total for each topic, not a list of who chose what.

Each fictional pupil gets one paper slip and marks one topic. No real class survey is needed for this lesson. The teacher counts the choices and uses the largest total; a tie goes to a class discussion.

Remove unnecessary fields before collection

A proposed form asks for name, full date of birth, home address, photograph and topic choice. For the stated counting task only the topic choice is needed. Remove the other fields before collection.

Do not fill them in and promise to ignore them later. If the **purpose** changes to individual attendance, design and justify that separate task with the teacher rather than silently reusing the survey.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A lunch helper needs food preferences and allergy information for its task, not a face photo, home address, or family income.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Follow the information from collection to disposal

> Name the audience, responsible person and agreed end point.

Trace the four-stage plan

A small information plan for a fictional vote

STAGE	NEEDED ACTION	PRIVACY CHECK
Collect	One topic per slip	No name or address
Count	Teacher totals choices	Keep slips restricted
Share	Three topic totals	No individual choices
Delete	After totals checked	Follow agreed plan

Purpose: choose a workshop topic. Collect the minimum, limit access, share the required result and agree how long the source slips are kept.

Check and share the needed totals

Suppose the fictional counts are drawing 8, music 7 and coding 5. These totals support the choice of drawing. There is no need to publish each slip or a photograph of a child holding it.

The teacher checks the total of 20 against the number of slips and then follows the agreed school **process** for disposing of the slips after verification.

Explain retention and deletion limits

- A **retention** plan needs a clear trigger and a responsible person: for example, the teacher disposes of the fictional vote slips after the totals are checked and the activity closes.

  Real classroom records must follow the school's approved **process**. Deleting one visible copy does not prove that every backup, screenshot or previously shared copy is gone.

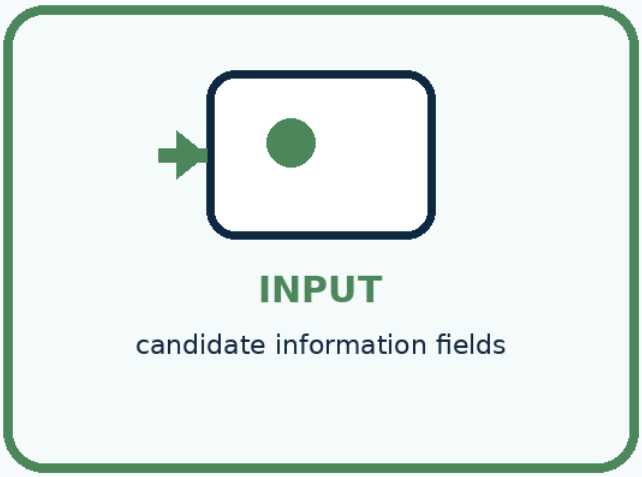
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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<summary>Earlier wording, alternatives and picture transcripts</summary>

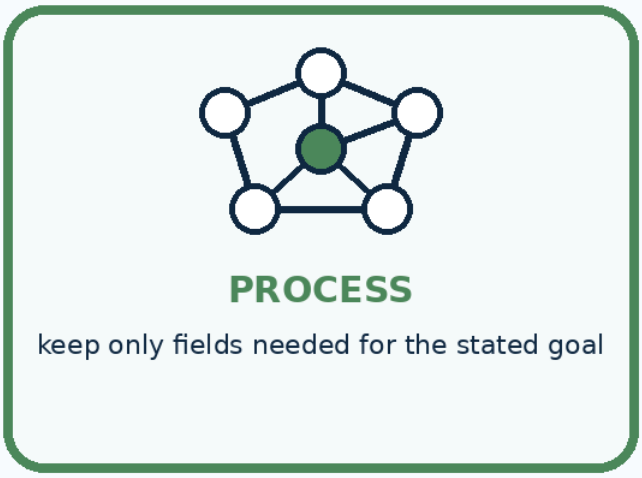
VISUAL **MECHANISM**

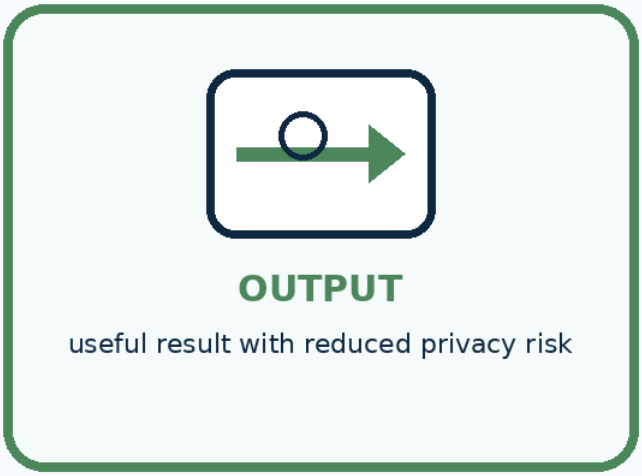
Input → Process → Output

Follow the information instead of imagining magic inside the machine.



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INPUT candidate information fields

PROCESS keep only fields needed for the stated goal

OUTPUT useful result with reduced **privacy** risk

CORE IDEA A useful system should collect only information it truly needs.

UNDER THE HOOD **Data minimization** limits collection, access, **retention**, and reuse to what is necessary for a defined **purpose**.

Less data reduces breach exposure and misuse, but essential high-stakes information still needs secure, consent-aware handling.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare less collection with hidden identifiers

> Removing a name and removing the need for identifying records are different decisions.

Pictures paired with their evidence



The case to investigate

CASE 1 — Allergy Clue: needed for safe lunch filter. Meaning: relevant sensitive data.



The case to investigate

CASE 2 — Favorite foods Clue: useful for preference. Meaning: optional relevant data.



The case to investigate

CASE 3 — Home address Clue: not needed. Meaning: remove.



The case to investigate

CASE 4 — Face photo Clue: not needed for menu choice. Meaning: remove.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY Useful does not mean permitted forever or for every new **purpose**.

▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Allergy Clue: needed for safe lunch filter. Meaning: relevant sensitive data.

CASE 2 — Favorite foods Clue: useful for preference. Meaning: optional relevant data.

CASE 3 — Home address Clue: not needed. Meaning: remove.

CASE 4 — Face photo Clue: not needed for menu choice. Meaning: remove.

Change one thing in the pictured case

Change: The teacher adds an authorised task to return one repaired tool to its borrower.

Prediction: A minimal contact **mechanism** may now be needed for that tool.

Reason: The **purpose** changed; reassess necessity rather than restoring every old field.

Trace a tool-repair count without borrower records

> Keep the original totals-only purpose separate from an authorised return task.

Read the purpose and proposed fields

A fictional teacher wants to count broken craft tools by type, without contacting borrowers.

Needed **output**: totals for scissors, ruler and hole-punch repairs.

Draft form requests pupil name, birthday, tool type and repair-needed Yes/No. No borrower follow-up is required.

Return to the workshop vote’s information journey

AI ACADEMY / AI-1/24 · 03 · INSPECT THE LESSON MODEL

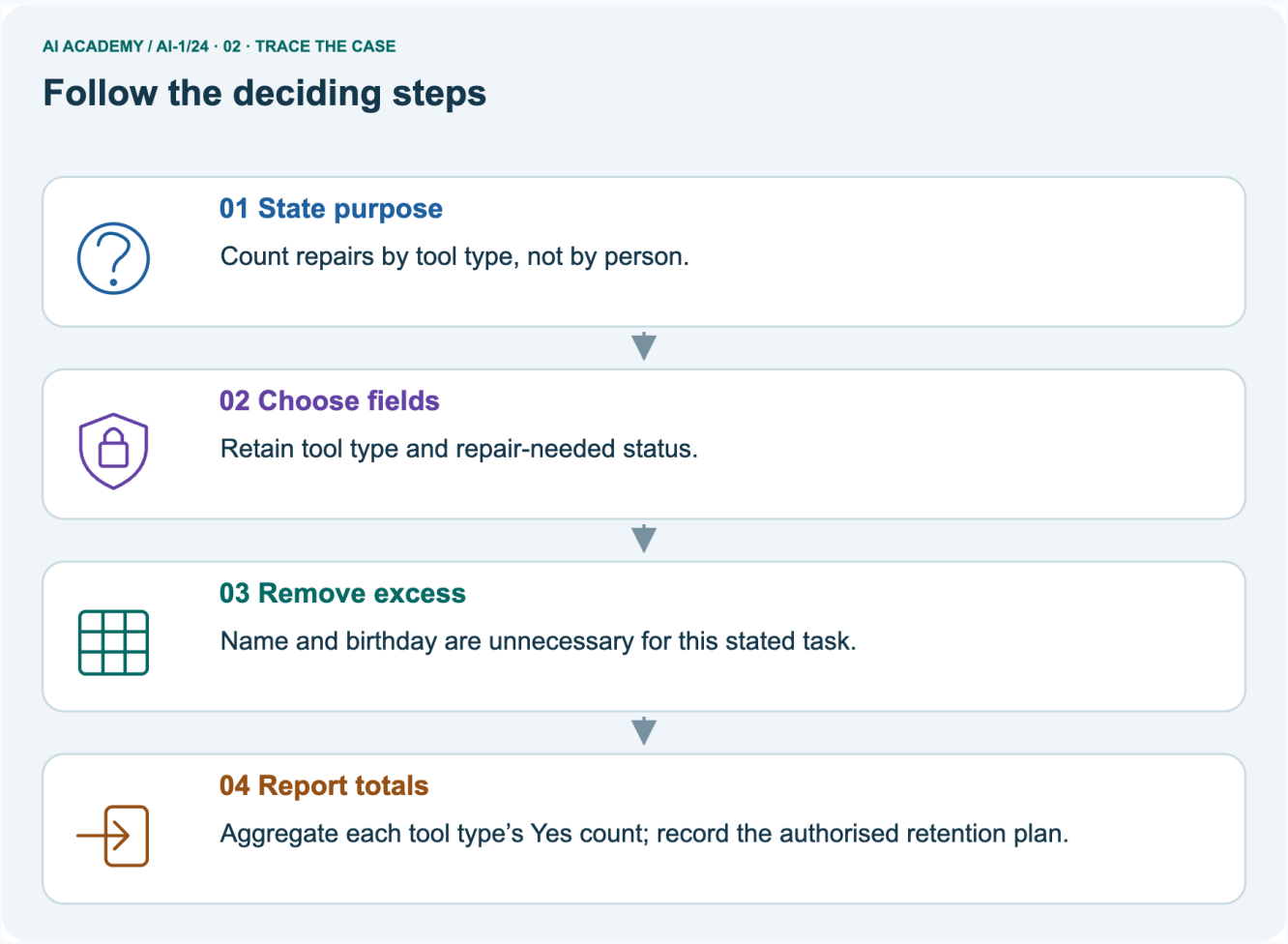
A small information plan for a fictional vote

Stage	Needed action	Privacy check
Collect	One topic per slip	No name or address
Count	Teacher totals choices	Keep slips restricted
Share	Three topic totals	No individual choices
Delete	After totals checked	Follow agreed plan

Purpose: choose a workshop topic. Collect the minimum, limit access, share the required result and agree how long the source slips are kept.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the smaller-input design



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State purpose	Count repairs by tool type, not by person.
2. Choose fields	Retain tool type and repair-needed status.
3. Remove excess	Name and birthday are unnecessary for this stated task.
4. Report totals	Aggregate each tool type’s Yes count; record the authorised retention plan.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain why each retained field is needed

- A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals.

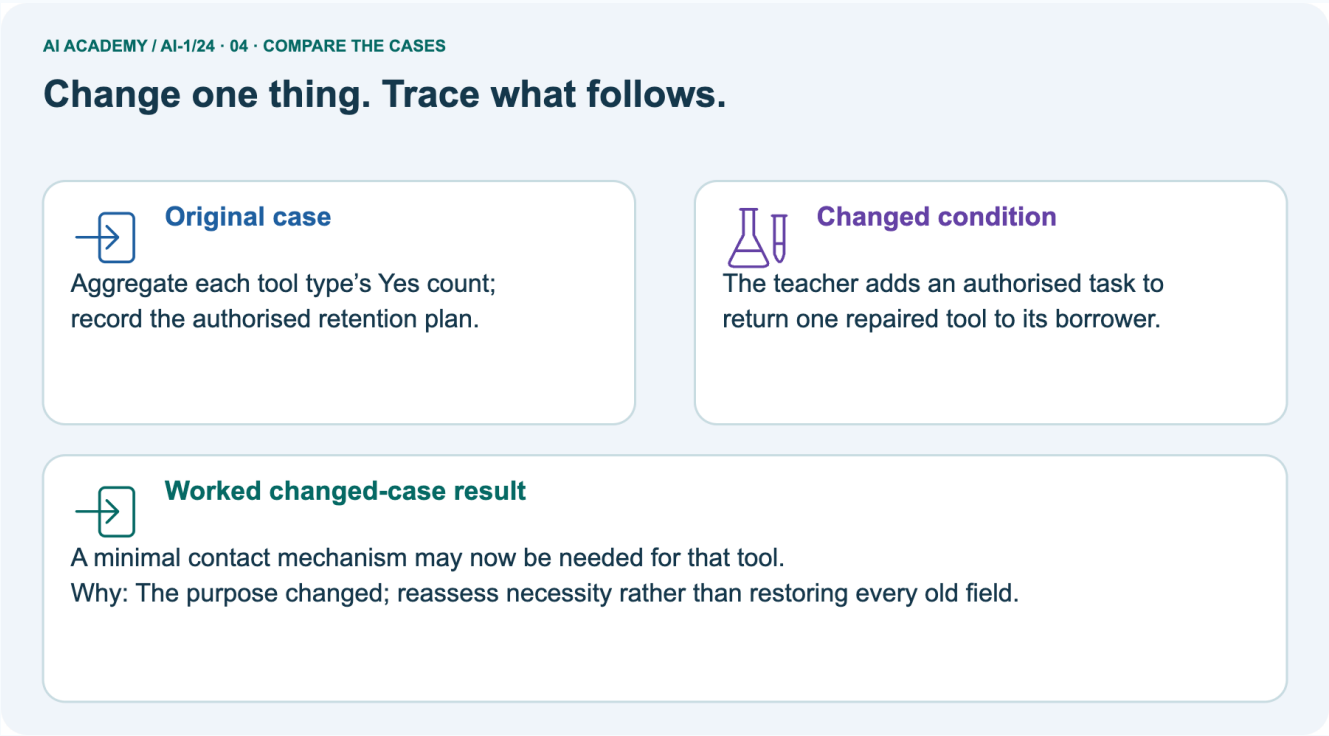
  Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

- Read the labelled picture: For the original task, does adding an anonymous-looking pupil code improve the repair count?

Predict a changed purpose

- Before opening the changed-purpose picture, suppose the teacher is authorised to return one repaired tool to its borrower. Which minimal contact mechanism could support that separate task? Which other old fields still have no justification?

▼ Check the changed-purpose design



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit anonymity claims

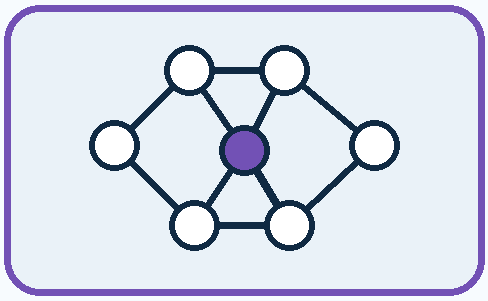
Removing these fields is not a universal guarantee of anonymity or permission to collect other identifying combinations.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A lunch helper needs food preferences and allergy information for its task, not a face photo, home address, or family income.

STEP 1 — OBSERVE candidate information fields

STEP 2 — TRACE keep only fields needed for the stated goal

STEP 3 — CONCLUDE useful result with reduced **privacy** risk

MODEL REASONING Keep only information required for safety and stated preferences, allow a person to provide or update it, restrict access, and delete it when the **purpose** ends.

Explain why each removed field adds risk without benefit.

DO NOT OVERCLAIM Useful does not mean permitted forever or for every new **purpose**.

What new **evidence** could change this answer?

A closer explanation

Proposed change	What it changes	What still needs thought
Replace a name with P07	Hides the name on this sheet	A separate key can reconnect the code
Remove names but keep photos	Removes one identifying field	Faces may still identify people

Proposed change	What it changes	What still needs thought
Share topic totals only	Removes individual rows	Small or unusual groups may still reveal a choice

A code is not a promise of anonymity. If a teacher keeps a key connecting P07 to a pupil, the coded record can still be linked to that pupil.

Even without a key, a combination such as an exact birthday, a team and a photograph may identify someone. Ask whether a reader could connect the information to a person, not just whether the name column is blank.

Totals also need context. A report saying that the only member of a tiny club chose music reveals that member’s choice.

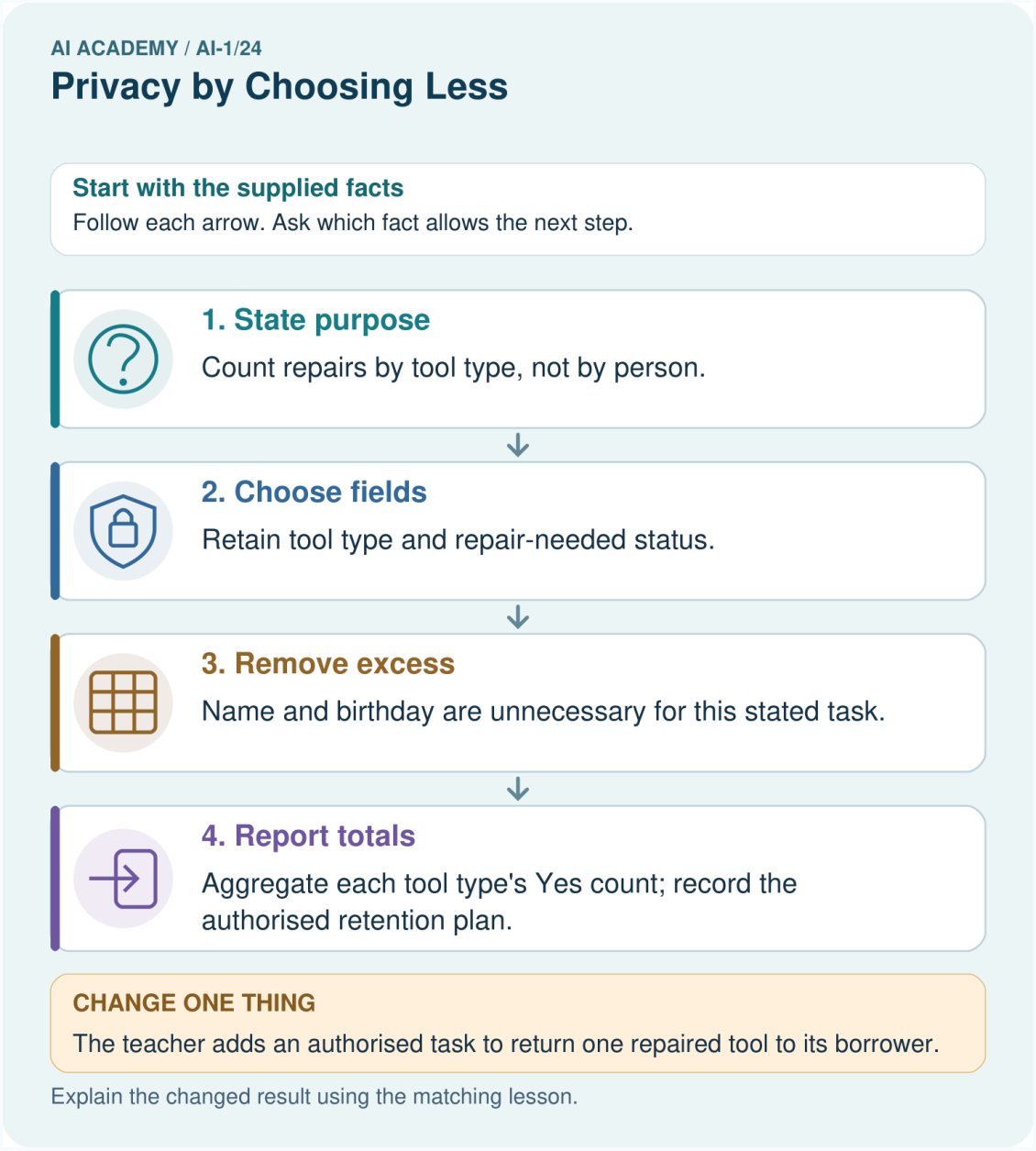
Our whole-class topic totals reveal less detail than individual rows, but we should not describe every possible total as automatically anonymous. Share only what the audience needs for the **purpose**.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Privacy by Choosing Less

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: The teacher adds an authorised task to return one repaired tool to its borrower. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Write a complete data plan and a sensible response

> A plan needs purpose, fields, access, sharing, retention and responsibility.

Use the planning questions

A useful data plan answers five questions: what is the **purpose**; which fields are necessary; who needs access; what result will be shared; and when will the source information be disposed of under the agreed **process**?

It also names who is responsible. “Keep everything safe forever” answers neither the collection question nor the **retention** question.

Respond to accidental sharing

If you accidentally share personal information, stop sharing it and tell the teacher or another trusted adult promptly. Do not repost it to ask classmates for help.

An adult can follow the school’s **process** and decide what action is appropriate. Avoid promising that deleting a message has removed every copy.

Repair the source’s tempting claims

- CLAIM A Collect data now in case it becomes useful later.
- CLAIM B Removing names makes all data anonymous and safe.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Allergy	needed for safe lunch filter	relevant sensitive data
Favorite foods	useful for preference	optional relevant data
Home address	not needed	remove
Face photo	not needed for menu choice	remove

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **For the fictional workshop vote, name one field to collect and two to remove. State the audience for the totals and a clear trigger for the teacher to dispose of the slips under the agreed process**

MAIN LESSON · SECTION 8

Practise: challenge a claim of anonymity

> A code, photograph or tiny-group total can still connect information to a person.

Use fictional inputs and approved tools

An AI tool is another place where information may be sent. Before using it for schoolwork, follow the teacher's approved tool and data rules.

Do not paste real names, contact details, photos, passwords or private messages into an unapproved service. For this lesson, use the provided fictional counts or work entirely on paper.

An AI tool is not needed to add 8 + 7 + 5.

Choose the audience before sharing

A student may need the lesson and their own work; a teacher may need restricted teaching records. Access should match these jobs. A hidden link or a hard-to-guess filename is not a substitute for an access check.

When information is shared more widely, it may be copied beyond the original audience. Decide the audience before sharing.

Challenge the coded-record claim in W2

Use workbook W2 for the connected example "**Privacy** and **data minimisation**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the more-data-is-always-better claim

- Use the worked case for **Privacy** and **data minimisation**. Proposed explanation: "More data always improves a task". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the tool-count question

For the original task, does adding an anonymous-looking pupil code improve the repair count?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A useful system should collect only information it truly needs.

Use a fresh example to explain **privacy**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Removing names makes all data anonymous and safe.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **A classmate proposes uploading the named vote spreadsheet to an AI tool to make a chart. The required chart shows only three topic totals. Propose a smaller input, explain what must be checked before tool use, and name a way to complete the task without uploading data**

Practise with less help: Challenge a claim of anonymity

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Redesign a fictional lunch-counting form

> Use a clear totals-only purpose and no real personal information.

The fictional task and full draft form

Paper redesign using fictional menu choices: a lunch helper only totals how many pupils select rice, pasta or soup. It makes no dietary-safety recommendation and does not identify or contact diners. The draft asks for name, birthday, home address, photograph, favourite

colour and menu choice. Use fictional counts rice=9, pasta=7, soup=4; do not collect real pupil or health information.

Predict, redesign and check the plan

1. Predict which fields the stated count requires, then cross out the rest and justify each decision. Check that the three totals sum to 20. Propose who counts the slips, who sees the totals, and the responsible person and agreed trigger for disposing of source slips. If the job changes to managing individual dietary safety, stop treating this totals-only plan as sufficient: the teacher must design that separate authorised process. Do not guess health needs from menu choices.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Record the result and purpose boundary

Case	Prediction	Observed result	What it means
Normal			
Edge			

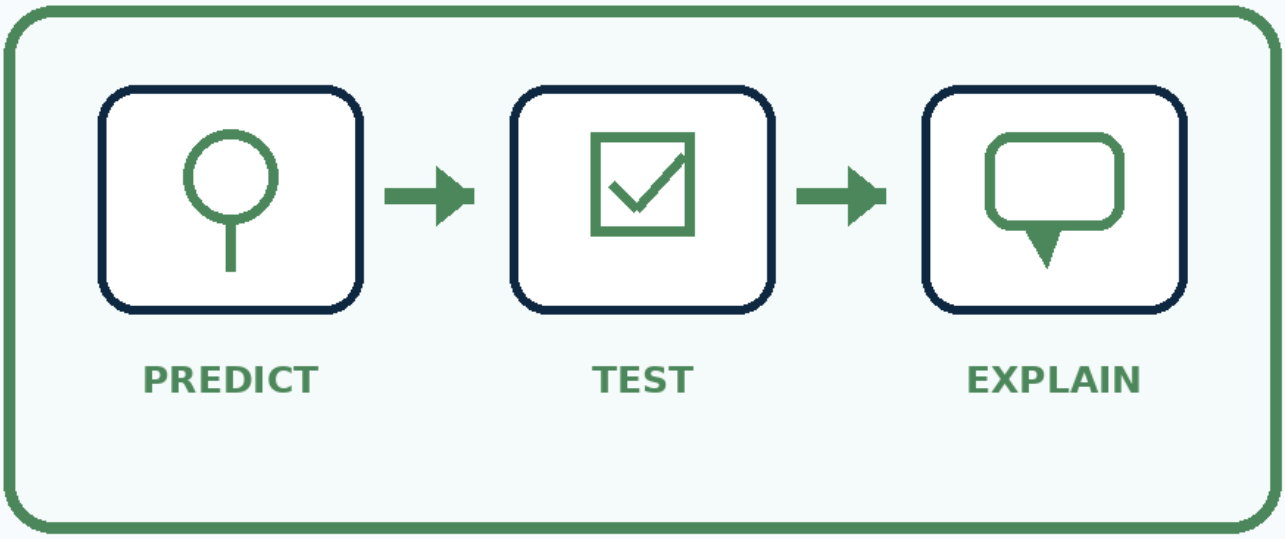
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Too-Much-Data Redesign

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Redesign a lunch helper that asks for too much **personal data**.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design the library's minimal-data task

> Use the new repair-category purpose before feedback.

The independent task and draft fields

A fictional library wants only the total number of books needing repair in each category: cover, page or binding. It does not need to contact borrowers.

A draft repair sheet asks for borrower name, home address, book repair category and borrower photograph.

Design a plan: state the **purpose**, required and removed fields, who handles source sheets, what totals are shared with the librarian, and who disposes of sheets at what agreed point. Use no real borrower details.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Design a different minimal-data task

> **My independent explanation, trace and limits**
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Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A book helper needs reading topic and approximate difficulty. Someone asks it to collect home address too. Which information can be left out, and why?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Cross out names, faces, and ownership details unless an explicit need survives review; set a deletion rule for sorter records.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A book recommender requests genre, maximum pages, home address and a selfie. Keep only fields needed for genre and length matching. Explain one consequence of storing extra fields.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Plan the sorter’s information before collecting it

> Justify fields and specify access and disposal for the actual classroom purpose.

This week’s project decision

THIS WEEK'S CHECKPOINT Cross out names, faces, and ownership details unless an explicit need survives review; set a deletion rule for sorter records.

Write the sorter’s **purpose**, necessary fields, access rules and deletion point before collecting any real classroom information.

Write the collection and access plan

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep future use within reviewed purposes

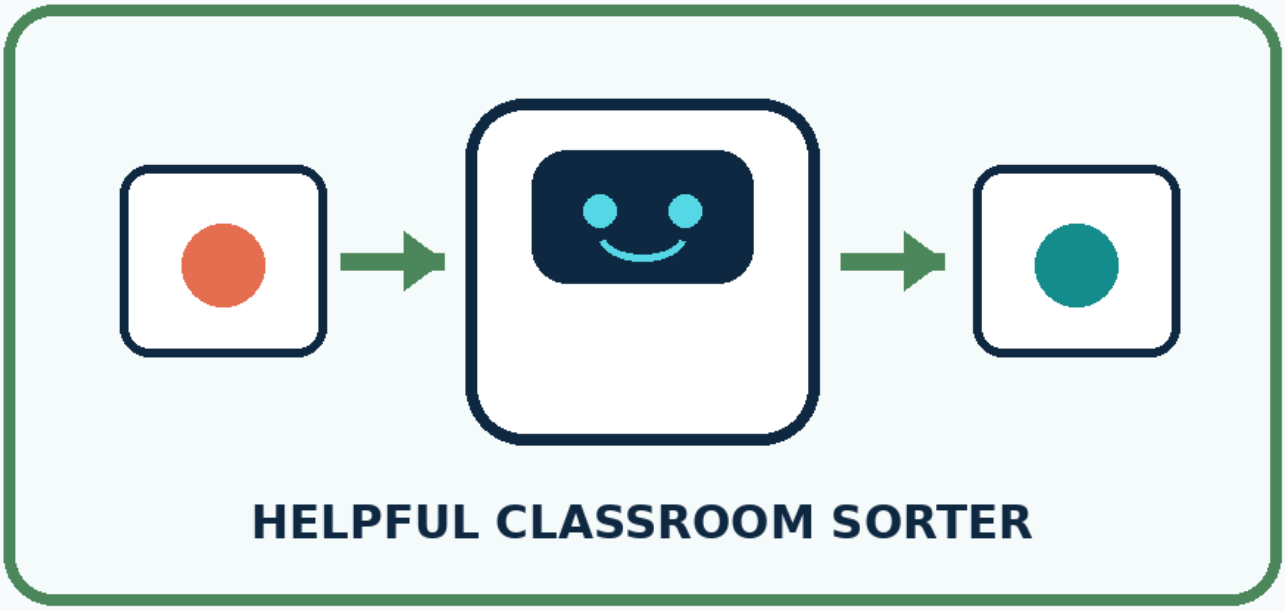
RESPONSIBLE DESIGN Useful does not mean permitted forever or for every new **purpose**.

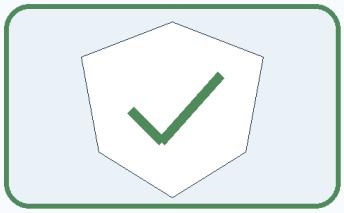
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify what you leave out

> Close your notes and connect every retained field to the purpose.

Your exit explanation

- Close your notes. Explain this goal in your own words: Justify which information a stated task needs and remove unnecessary fields

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
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Choose a changed-case check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
privacy	control and protection of information about people	
data minimization	collecting only what the purpose requires	
purpose	specific reason information is used	
retention	how long information is kept	

Terms for the connected explanation

Word	Meaning
Personal data	Information about an identifiable person, including combinations that can identify someone
Data minimisation	Using only the information needed for a stated purpose
Access control	A rule and mechanism deciding who can see or use information

Repair a claim using a word

- CLAIM TO REPAIR Collect data now in case it becomes useful later.
- Use one vocabulary word and one piece of **evidence** to repair it:

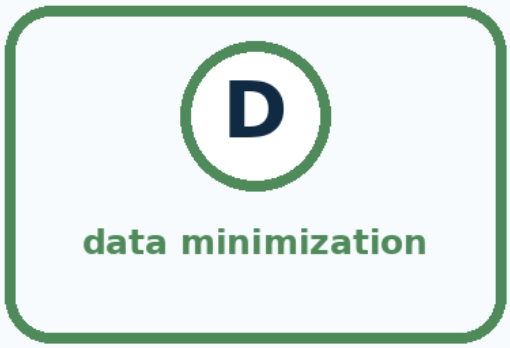
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<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





</details>





Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

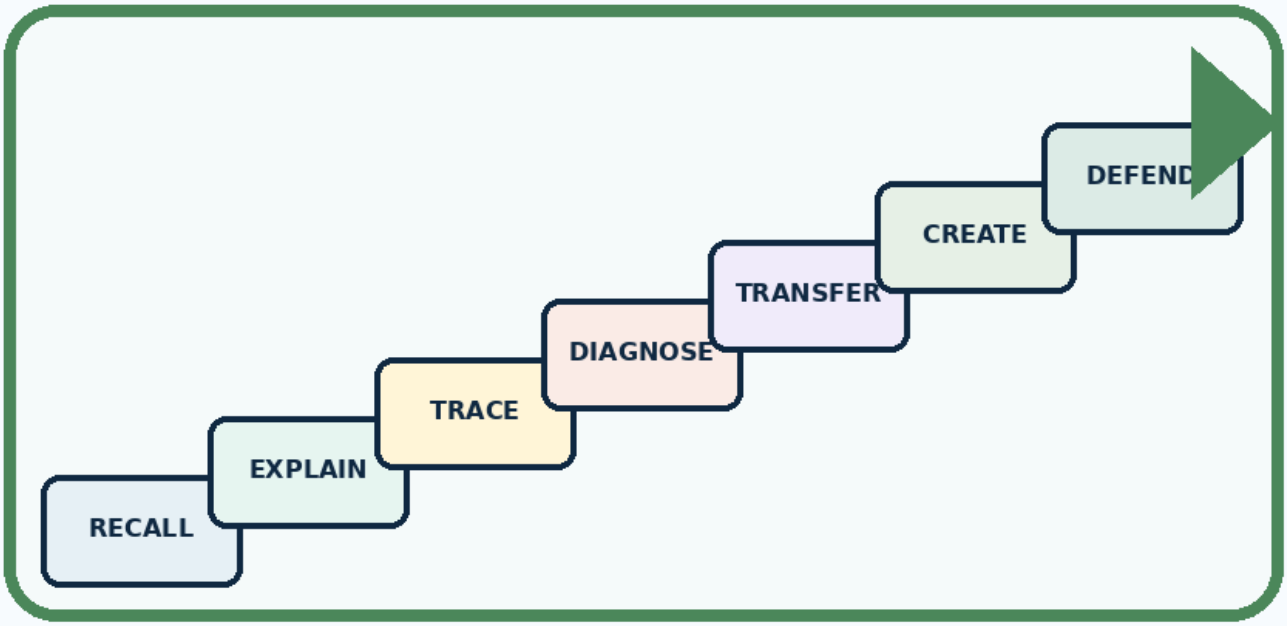
- RECALL Define **privacy**.
- EXPLAIN A useful system should collect only information it truly needs.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Collect data now in case it becomes useful later.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Design a different minimal-data task is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/**privacy** concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Design a different minimal-data task is the selected independent case.



EXPLANATION CHECK Justify necessary fields, access, shared output and responsible disposal for the stated purpose.

Independent **evidence**:

SELF-CHECK Keep only information required for safety and stated preferences, allow a person to provide or update it, restrict access, and delete it when the **purpose** ends.

Explain why each removed field adds risk without benefit.

NEXT BRIDGE Carry today's **evidence** habit into: Generate, Retrieve, or Calculate?.

▼ **Extra practice: Collect only what the book task needs**

REFERENCE · SECTION 13

Extra practice: Collect only what the book task needs

> Try the question before opening its worked explanation.

Read the supplied case and question

A genre-based book helper asks for genre, home address and a photograph.

Which fields are necessary?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

Evidence or stage	Value or result
Genre	Useful for genre matching
Home address	Not needed for this task
Photograph	Not needed for this task

▼ Earlier wording, alternatives and picture transcripts

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Collect only what the book task needs

A genre-based book helper asks for genre, home address and a photograph.

Which fields are necessary?

Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Justify which information a stated task needs and remove unnecessary fields	Explain the deciding step, show a trace or test, and state one limit.
Distinguish reducing personal data from hiding a name	Explain the deciding step, show a trace or test, and state one limit.
Design a simple access, retention and deletion plan for a classroom activity	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional teacher wants to count broken craft tools by type, without contacting borrowers.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional teacher wants to count broken craft tools by type, without contacting borrowers.
- Which labelled step matches this detail? Aggregate each tool type's Yes count; record the authorised **retention** plan.
- 1. State **purpose**
- 2. Choose fields
- 3. Remove excess
- 4. Report totals
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Report totals Aggregate each tool type's Yes count; record the authorised **retention** plan.

A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals.

Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

Changed-case question

- Changed-case practice: The teacher adds an authorised task to return one repaired tool to its borrower. Which conclusion is supported by the supplied facts?
- 1. A minimal contact **mechanism** may now be needed for that tool.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: A minimal contact **mechanism** may now be needed for that tool. The **purpose** changed; reassess necessity rather than restoring every old field.

Repair and save your work

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, unit and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Review Confidence Is Not Correctness (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

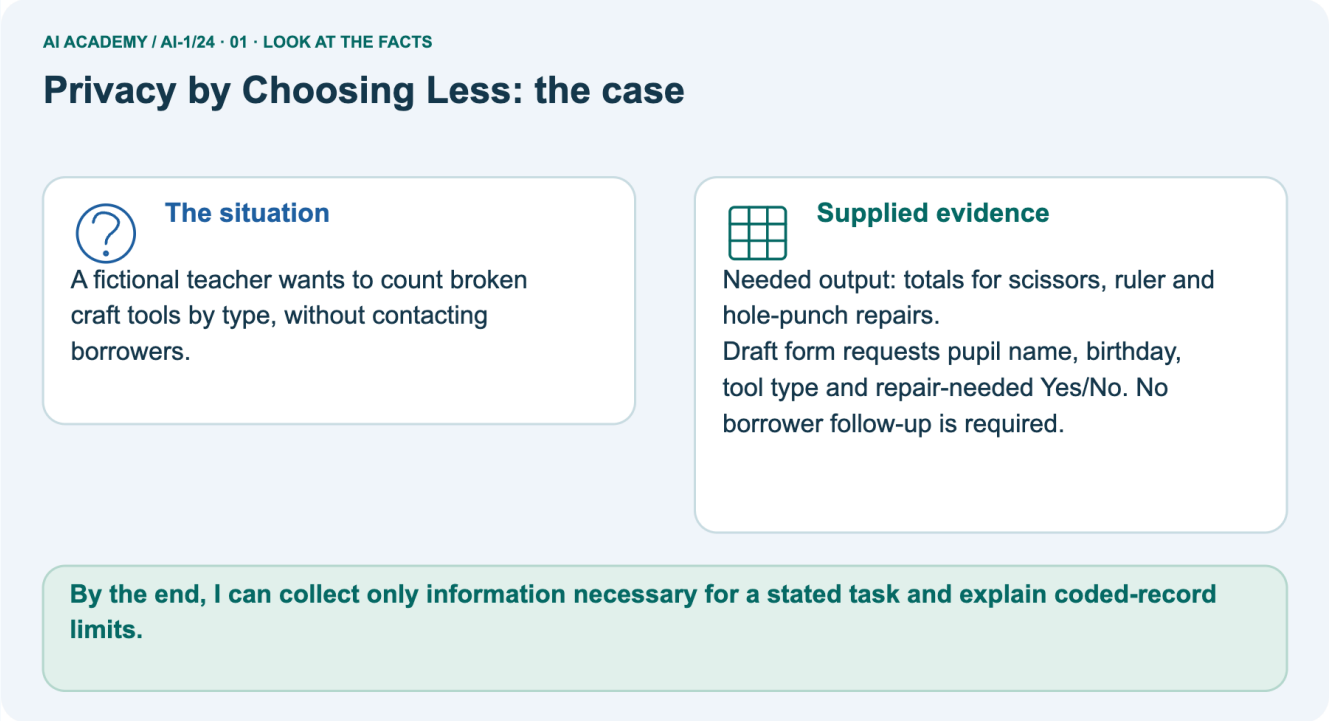
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Privacy by Choosing Less: the case



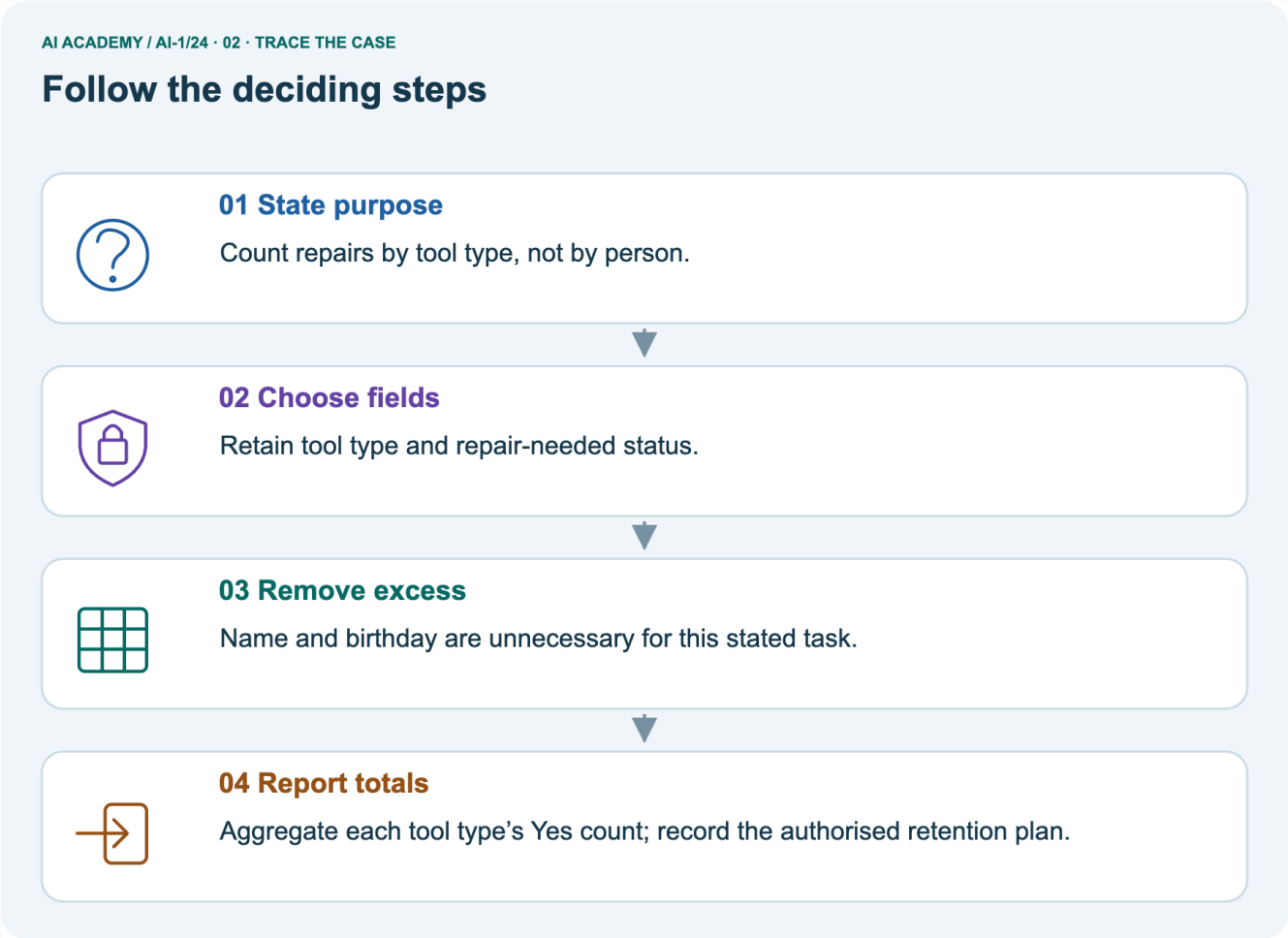
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional teacher wants to count broken craft tools by type, without contacting borrowers.
Supplied facts: Needed **output**: totals for scissors, ruler and hole-punch repairs.

Draft form requests pupil name, birthday, tool type and repair-needed Yes/No. No borrower follow-up is required.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

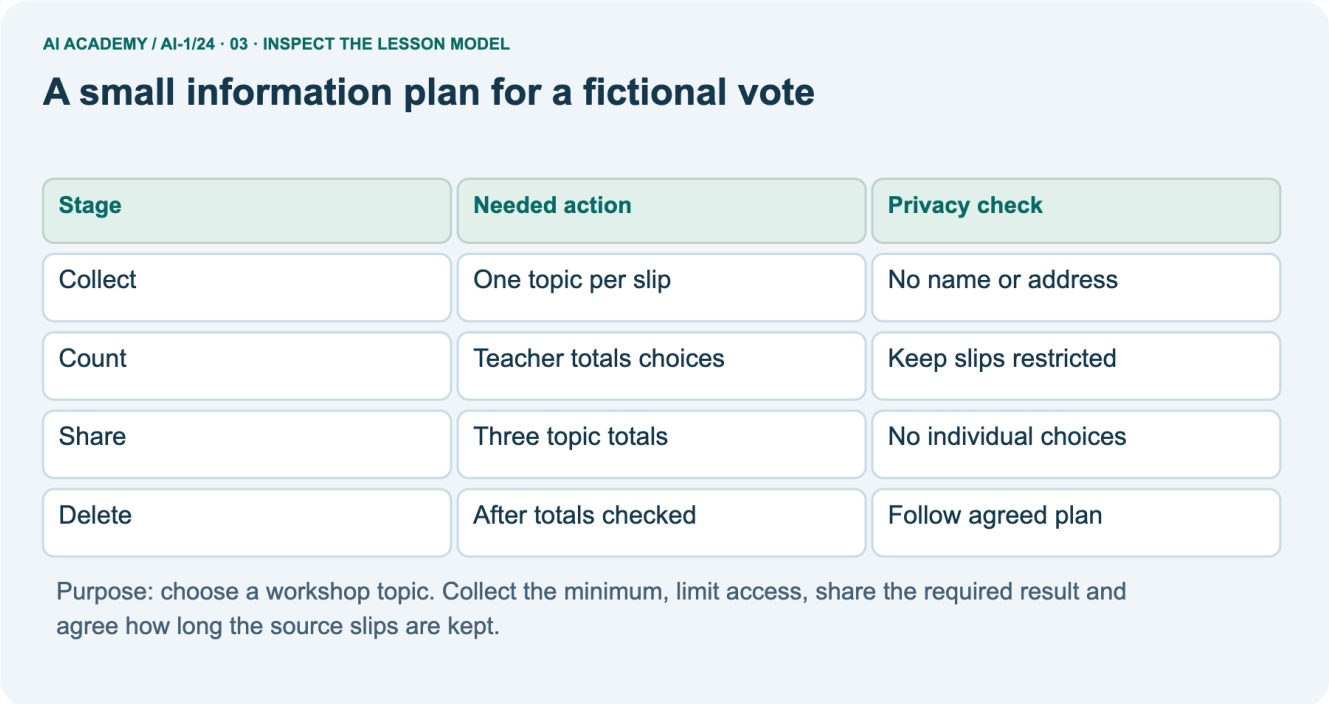


Follow the labelled picture with the matching explanation. Open the vector diagram

State **purpose**: Count repairs by tool type, not by person. → Choose fields: Retain tool type and repair-needed status. → Remove excess: Name and birthday are unnecessary for this stated task. → Report totals: Aggregate each tool type's Yes count; record the authorised **retention** plan.

Point to each numbered stage and explain which recorded **input** it uses.

03. A small information plan for a fictional vote



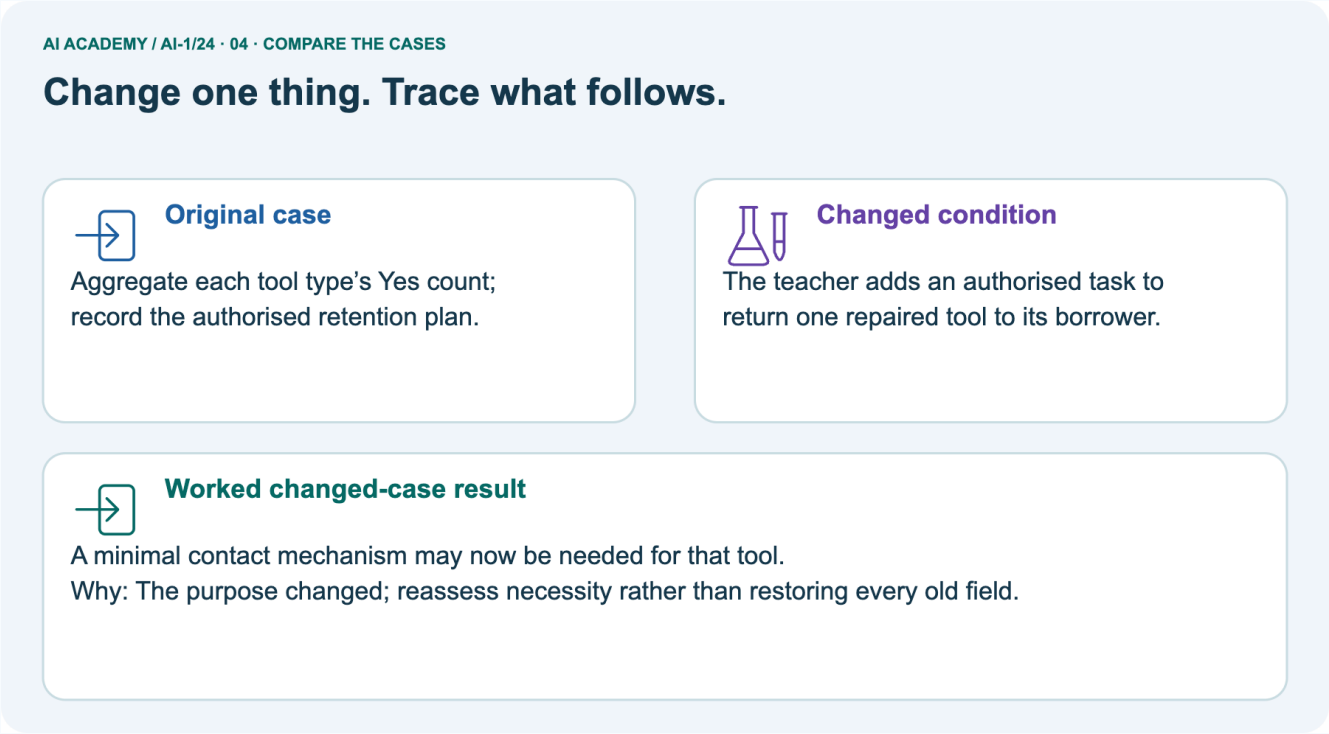
Follow the labelled picture with the matching explanation. Open the vector diagram

A small information plan for a fictional vote. Collect | One topic per slip | No name or address. Count | Teacher totals choices | Keep slips restricted. Share | Three topic totals | No individual choices.

Delete | After totals checked | Follow agreed plan

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



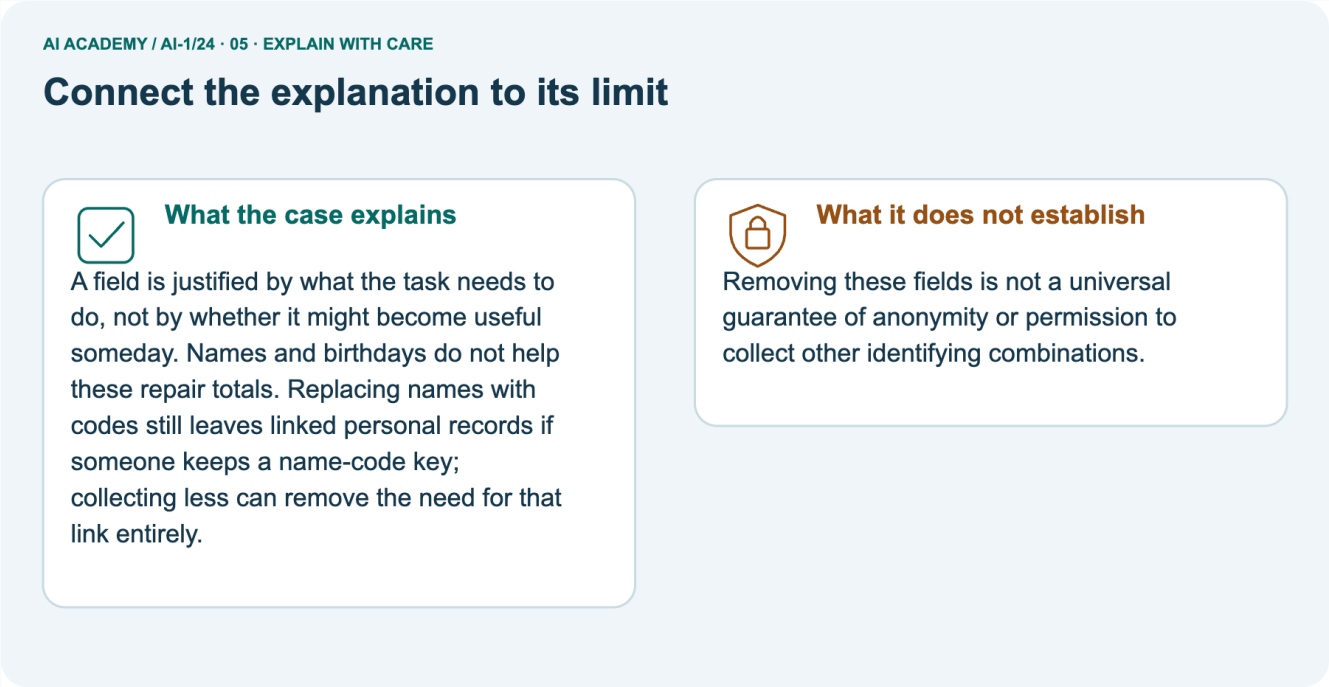
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Aggregate each tool type's Yes count; record the authorised **retention** plan. Changed condition: The teacher adds an authorised task to return one repaired tool to its borrower.

Expected result: A minimal contact **mechanism** may now be needed for that tool. Reason: The **purpose** changed; reassess necessity rather than restoring every old field.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

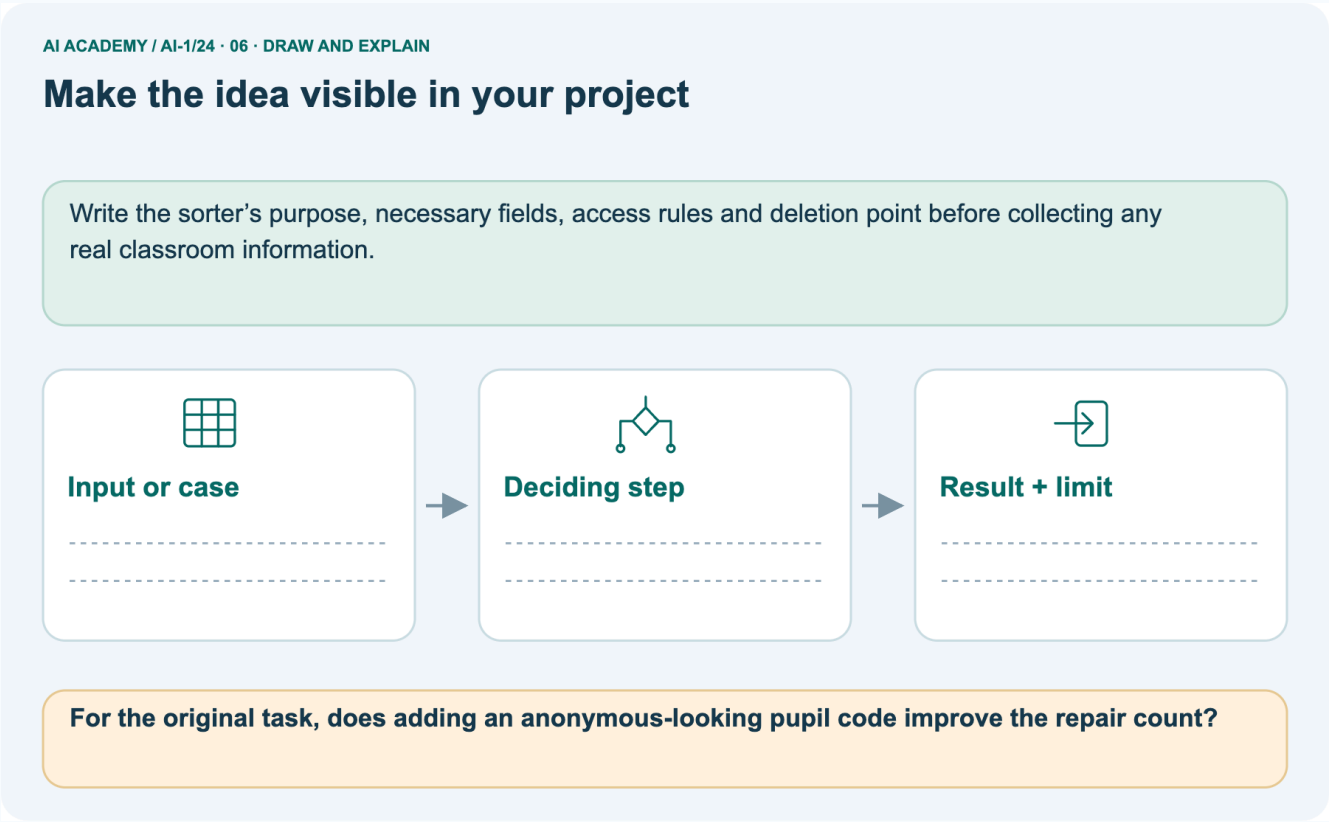
A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals.

Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

Limit: Removing these fields is not a universal guarantee of anonymity or permission to collect other identifying combinations.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write the sorter's **purpose**, necessary fields, access rules and deletion point before collecting any real classroom information. Draw your **input**, deciding step, result and limit.

Supported question: For the original task, does adding an anonymous-looking pupil code improve the repair count?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Privacy by Choosing Less: the case

02. Follow the deciding steps

03. A small information plan for a fictional vote

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.